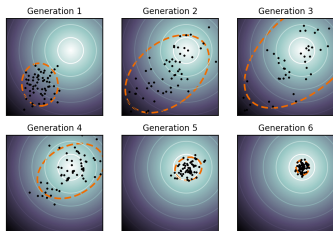
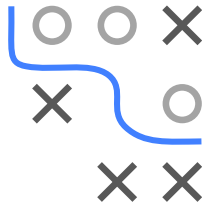


Optimization in Machine Learning

Evolutionary Algorithms

CMA-ES Algorithm



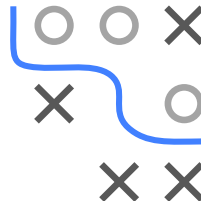
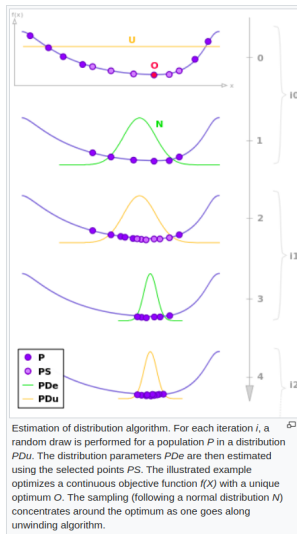
Learning goals

- CMA-ES strategy
- Estimation of distribution
- Step size control

ALGORITHM

Instead of population, maintain distribution
to sample offspring from

- ➊ Draw λ offspring $\mathbf{x}^{(i)}$ from $p(\cdot|\boldsymbol{\theta}^{[t]})$
- ➋ Evaluate fitness $f(\mathbf{x}^{(i)})$
- ➌ Update $\boldsymbol{\theta}^{[t+1]}$ with μ best offspring

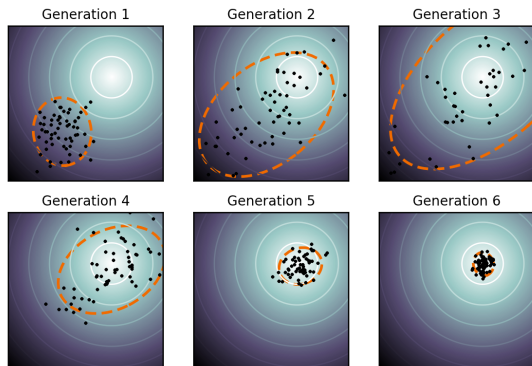


COVARIANCE MATRIX ADAPTATION

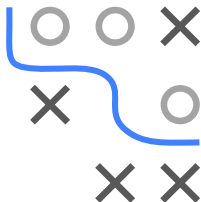
- Sample distribution is multivariate Gaussian

$$\mathbf{x}^{[t+1](i)} \sim \mathbf{m}^{[t]} + \sigma^{[t]} \mathcal{N}(\mathbf{0}, \mathbf{C}^{[t]}) \quad \text{for } i = 1, \dots, \lambda$$

- $\mathbf{x}^{[t+1](i)} \in \mathbb{R}^d$ i -th offspring; $\lambda \geq 2$ number of offspring
- $\mathbf{m}^{[t]} \in \mathbb{R}^d$ mean value and $\mathbf{C}^{[t]} \in \mathbb{R}^{d \times d}$ covariance matrix
- $\sigma^{[t]} \in \mathbb{R}_+$ “overall” standard deviation/step size

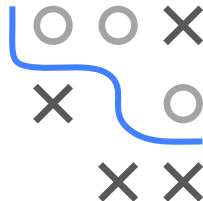


- **Question:** How to adapt $\mathbf{m}^{[t+1]}$, $\mathbf{C}^{[t+1]}$, $\sigma^{[t+1]}$ for next generation $t + 1$?



CMA-ES: BASIC METHOD - ITERATION 1

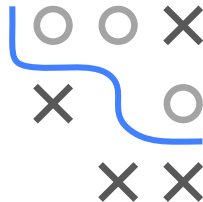
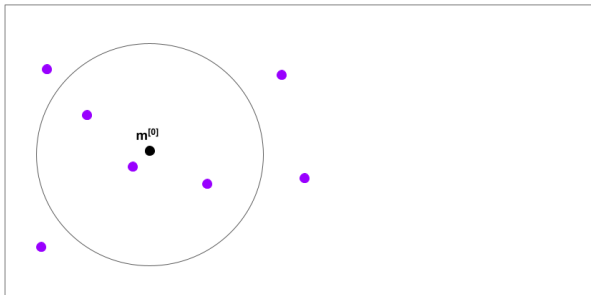
- 1 Initialize $\mathbf{m}^{[0]}$, $\sigma^{[0]}$ problem-dependent and $\mathbf{C}^{[0]} = \mathbf{I}_d$



CMA-ES: BASIC METHOD - ITERATION 1

- ❶ **Sample** λ offspring from distribution

$$\mathbf{x}^{[1](i)} = \mathbf{m}^{[0]} + \sigma^{[0]} \mathcal{N}(\mathbf{0}, \mathbf{C}^{[0]})$$

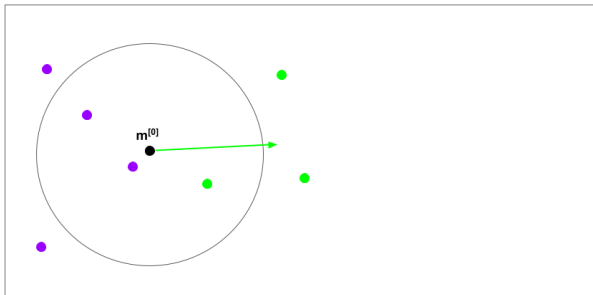
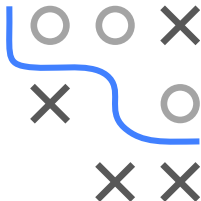


CMA-ES: BASIC METHOD - ITERATION 1

- ② **Selection and recombination** of $\mu < \lambda$ best-performing offspring using fixed weights $w_1 \geq \dots \geq w_\mu > 0$, $\sum_{i=1}^{\mu} w_i = 1$. $\mathbf{x}_{i:\lambda}$ is i -th ranked solution, ranked by $f(\mathbf{x}_{i:\lambda})$.

Calculation of auxiliary variables ($\mu = 3$ points)

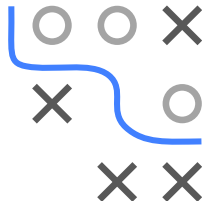
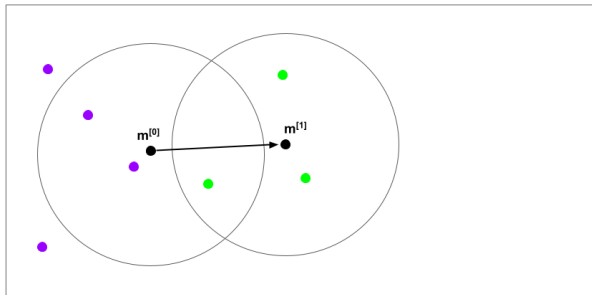
$$\mathbf{y}_w^{[1]} := \sum_{i=1}^{\mu} w_i (\mathbf{x}_{i:\lambda}^{[1]} - \mathbf{m}^{[0]}) / \sigma^{[0]} := \sum_{i=1}^{\mu} w_i \mathbf{y}_{i:\lambda}^{[1]}$$



CMA-ES: BASIC METHOD - ITERATION 1

3 Update mean

Movement towards the new distribution with mean $\mathbf{m}^{[1]} = \mathbf{m}^{[0]} + \sigma^{[0]} \mathbf{y}_w^{[1]}$.



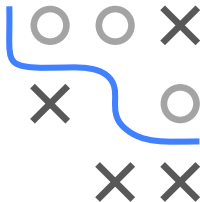
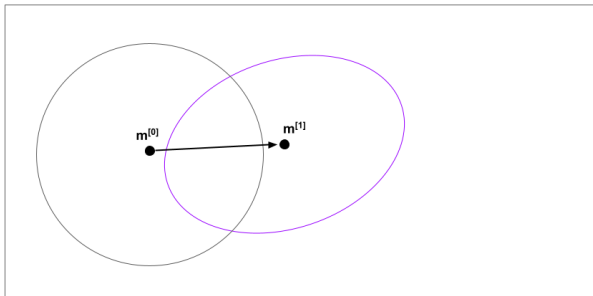
CMA-ES: BASIC METHOD - ITERATION 1

- 3 Update covariance matrix** Roughly: elongate density ellipsoid in direction of successful steps.
- $\mathbf{C}^{[1]}$ reproduces successful points with higher probability than $\mathbf{C}^{[0]}$.
 - Update $\mathbf{C}^{[0]}$ using sum of outer products and parameter c_μ :

$$\mathbf{C}^{[1]} = (1 - c_\mu)\mathbf{C}^{[0]} + c_\mu \sum_{i=1}^{\mu} w_i \mathbf{y}_{i;\lambda}^{[1]} (\mathbf{y}_{i;\lambda}^{[1]})^T \text{ (rank-}\mu \text{ update).}$$

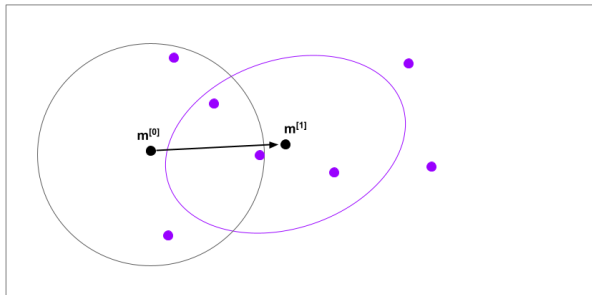
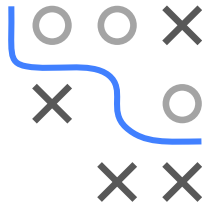
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 (rank- μ update).



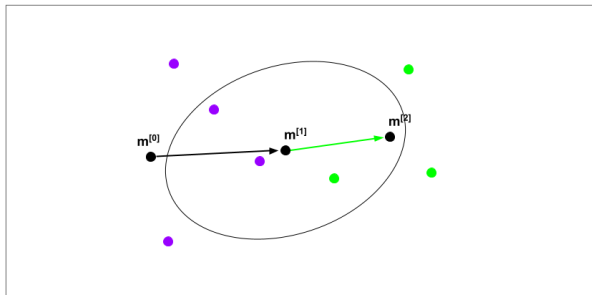
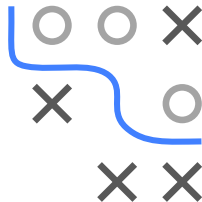
CMA-ES: BASIC METHOD - ITERATION 2

- 1 **Sample** from distribution for new generation



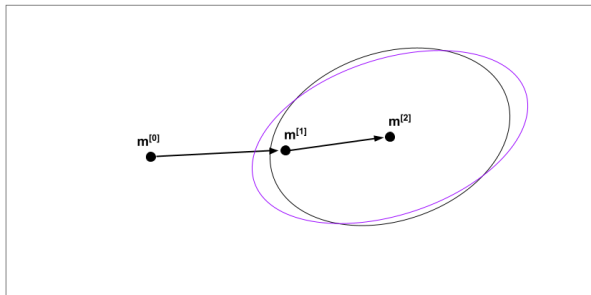
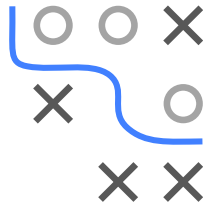
CMA-ES: BASIC METHOD - ITERATION 2

- 2 Selection and recombination of $\mu < \lambda$ best-performing offspring
- 3 Update mean



CMA-ES: BASIC METHOD - ITERATION 2

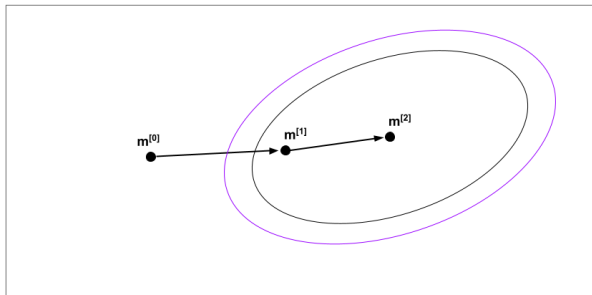
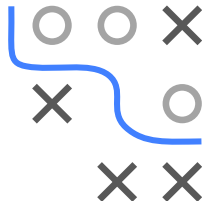
3 Update covariance matrix



CMA-ES: BASIC METHOD - ITERATION 2

③ Update step-size exploiting correlation in history of steps:

- steps point in similar direction $\implies$ increase step-size
- steps cancel out $\implies$ decrease step-size



UPDATING C: FULL UPDATE

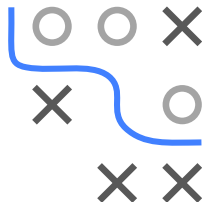
Full CMA update of $\mathbf{C}$ combines rank- μ update with a rank-1 update using exponentially smoothed evolution path $\mathbf{p}_c \in \mathbb{R}^d$ of successive steps and learning rate c_1 :

$$\mathbf{p}_c^{[0]} = \mathbf{0}, \quad \mathbf{p}_c^{[t+1]} = (1 - c_1)\mathbf{p}_c^{[t]} + \sqrt{\frac{c_1(2 - c_1)}{\sum_{i=1}^{\mu} w_i^2}} \mathbf{y}_w$$

Final update of $\mathbf{C}$ is

$$\mathbf{C}^{[t+1]} = (1 - c_1 - c_{\mu} \sum_j w_j) \mathbf{C}^{[t]} + c_1 \underbrace{\mathbf{p}_c^{[t+1]} (\mathbf{p}_c^{[t+1]})^T}_{\text{rank-1}} + c_{\mu} \underbrace{\sum_{i=1}^{\mu} w_i \mathbf{y}_{i:\lambda}^{[t+1]} (\mathbf{y}_{i:\lambda}^{[t+1]})^T}_{\text{rank-}\mu}$$

- Correlation between generations used in rank-1 update
- Information from entire population is used in rank- μ update



UPDATING σ : METHODS STEP-SIZE CONTROL

- **1/5-th success rule:** increases the step-size if more than 20 % of the new solutions are successful, decrease otherwise
- **σ -self-adaptation:** mutation is applied to the step-size and the better - according to the objective function value - is selected
- **Path length control via cumulative step-size adaptation (CSA)**
Intuition:
 - Short cumulative step-size $\triangleq$ steps cancel $\rightarrow$ decrease $\sigma^{[t+1]}$
 - Long cumulative step-size $\triangleq$ corr. steps $\rightarrow$ increase $\sigma^{[t+1]}$

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- Short cumulative step-size $\triangleq$ steps cancel $\rightarrow$ decrease $\sigma^{[t+1]}$
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